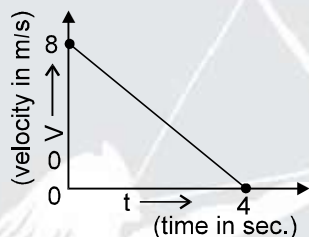


1. A block of mass 2 kg is given a push for a moment horizontally and then the block starts sliding over a horizontal plane. The graph shows the velocity-time graph of the motion. The co-efficient of sliding friction between the plane and the block is:



- (A) 0.02 (B) 0.20 (C) 0.04 (D) 0.40
2. Consider a uniformly charged solid non conducting sphere of radius R and charge Q . Consider three spherical regions centered at the centre of the uniformly charged solid non conducting sphere, as given below :

Region (1) $R_1 \geq r \geq 0$

Region (2) $R_2 \geq r \geq R_1$

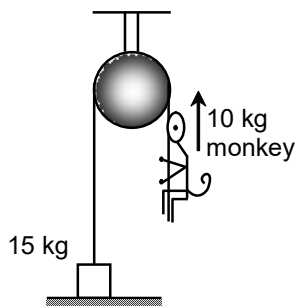
Region (3) $\infty \geq r \geq R_2$

If electrostatic potential energy stored in each of the three regions is equal then $\frac{R_2}{R_1}$ is :

- (A) 2 (B) 3 (C) 1 (D) 4
3. A plane transverse wave propagates along positive x -axis in a homogeneous medium density 200 kg/m^3 due to propagation of wave, medium particles oscillates and the oscillation energy density of particles in medium is $0.16 \pi^2 \text{ J/m}^3$ and maximum shear strain produced in medium is $8\pi \times 10^{-5}$. If the phase difference between two particles located at $x = 1\text{m}$ and $x = 2\text{m}$ is 144° . The most appropriate equation of wave will be

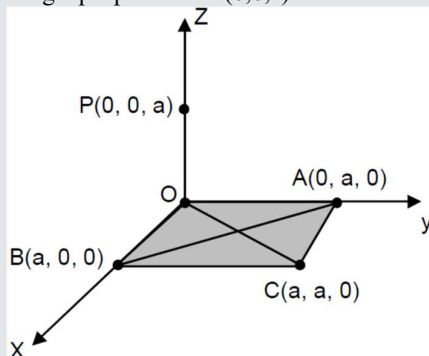
- (A) $y = 2 \times 10^{-4} \sin[200\pi t - 0.4\pi x]$ (B) $y = 5 \times 10^{-2} \sin[200\pi t - 0.8\pi x]$
(C) $y = 10^{-4} \sin[4007\pi t - 0.8\pi x]$ (D) $y = 10^{-2} \sin[200\pi t - 0.47\pi x]$

4. A monkey weighing 10 kg is climbing a massless rope and frictionless pulley attached to a 15 kg mass at the end as shown in fig. In order to raise the 15 kg mass of the ground, the monkey must climb-



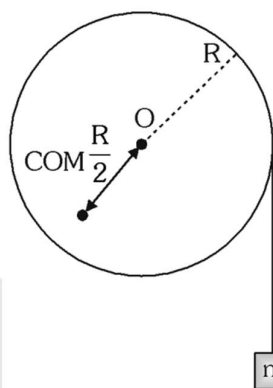
- (A) with a high uniform velocity (B) with a uniform acceleration g
 (C) with an acceleration greater than $\frac{g}{2}$ (D) with an acceleration greater than $\frac{g}{4}$

5. A point charge q is placed at $P(0,0,a)$. Then select correct alternative/s.



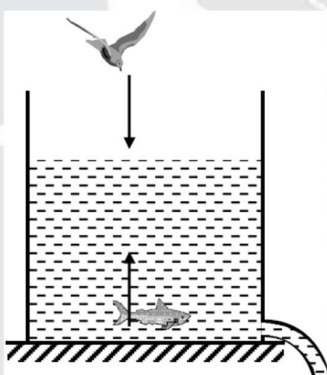
- (A) $\frac{q}{24\epsilon_0} > \phi_{OAB} > \frac{q}{48\epsilon_0}$ (B) $\phi_{OAC} = \frac{q}{48\epsilon_0}$
 (C) $\phi_{ABC} < \phi_{ABO}$ (D) $\phi_{OAC} = \phi_{OBC}$
6. A string of length 120 cm is clamped at both ends. By a tuning fork, oscillation are induced in string for which sustained stationary waves are produced in it. In the string it is observed that all points on string which have amplitude 3.5 mm are separated by 15 cm length on string. The amplitude of antinode is
 (A) $3.5\sqrt{2}$ mm (B) 3.5 mm (C) $7\sqrt{2}$ mm (D) 7 mm
7. One end-of a thin long uniform string having linear mass density λ is fixed to a wall and other end with a block which can move along a vertical rod. The string is horizontal and has a tension T_0 . If block executes SHM along the rod with vertical small amplitude A and angular frequency ω . The energy transferred to string in one oscillation of block is
 (A) $\sqrt{3T_0\lambda} A^2\omega\pi$ (B) $\sqrt{2T_0\lambda} A^2\omega\pi$ (C) $\sqrt{T_0\lambda} A^2\omega\pi$ (D) $\sqrt{5T_0\lambda} A^2\omega\pi$
8. A non-uniform disc and a block is shown in figure. The COM of disc is located at a distance $\frac{R}{2}$ from its geometrical centre O. The disc is pivoted at O and can spin freely about the point O in a vertical plane. A light thread is wrapped tightly over the disc and a small block of mass m is attached to its free end.

Mass of disc is four times the mass of block. Moment of inertia of disc about the axis through O is I. The block is pulled down a little from its equilibrium. The time period for very small oscillation is



- (A) $T = 2\pi \sqrt{\frac{I + mR^2}{2\sqrt{5}mgR}}$ (B) $T = 2\pi \sqrt{\frac{I + mR^2}{\sqrt{5}mgR}}$ (C) $T = 2\pi \sqrt{\frac{I + mR^2}{\sqrt{2}mgR}}$ (D) $T = 2\pi \sqrt{\frac{I + mR^2}{\sqrt{3}mgR}}$

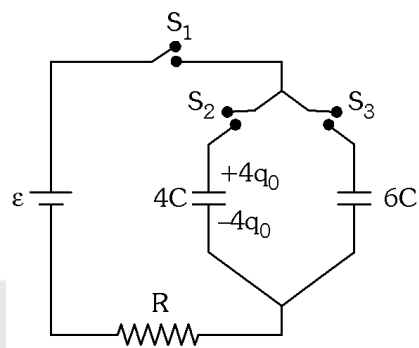
9. A bird in air is diving vertically downward over a tank with speed 5 cm/s while base of tank is silvered. A fish in the tank is rising upward along the same line with speed 2 cm/s. Water level is falling at rate of 2 cm/s. Take $\mu_{\text{water}} = 4/3$.



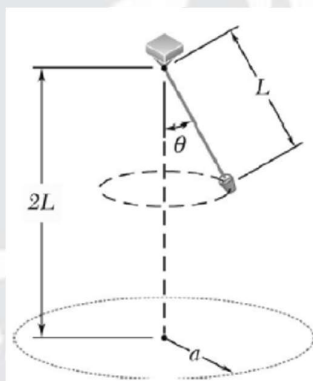
- (A) Speed of image of fish as seen by bird directly is 6 cm/s.
 (B) Speed of image of fish as seen by bird directly is 3 cm/s.
 (C) Speed of image of fish as seen by bird after reflection from mirror is 3 cm/s.
 (D) Speed of image of fish as seen by bird after reflection from mirror is 6 cm/s.

10. Two capacitors, one resistor and an ideal battery are connected through switches S_1 , S_2 and S_3 as shown in figure. Initially capacitor of capacitance $4C$ is charged, while other is uncharged. The upper

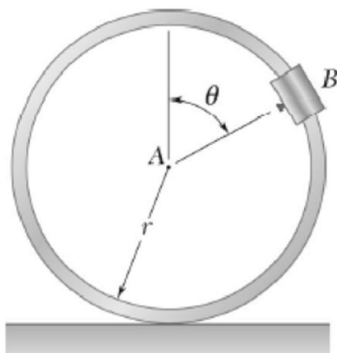
plate of capacitor $4C$ is positive plate. Now, switches S_1 , S_2 and S_3 are closed simultaneously then answer the following questions



- (A) The current resistor 'R' in clockwise direction as a function of time is $\left(\frac{5C\varepsilon - 2q_0}{5RC}\right)e^{-\frac{t}{10RC}}$
- (B) The current resistor 'R' in clockwise direction as a function of time is $\left(\frac{5C\varepsilon - 2q_0}{4RC}\right)e^{-\frac{t}{10RC}}$
- (C) The charge on the capacitor of capacitance '6C' as a function of time is $6C\varepsilon - \left(6C\varepsilon + \frac{12}{5}q_0\right)e^{-\frac{t}{10RC}}$
- (D) The charge on the capacitor of capacitance '6C' as a function of time is $6C\varepsilon - \left(6C\varepsilon - \frac{12}{5}q_0\right)e^{-\frac{t}{10RC}}$
11. A bucket is attached to a rope of length $L = 1.2$ m and is made to revolve in a horizontal circle. Drops of water leaking from the bucket fall and strike the floor along the perimeter of a circle of radius a ($\theta = 30^\circ$) (Take $g = 10$ m/s²)



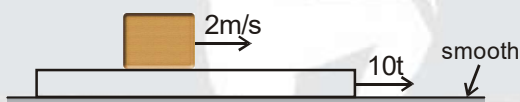
- (A) Time of flight of drop is 0.52 sec.
- (B) Time of flight of drop is 3.15 sec.
- (C) Radius A is 1.14 m
- (D) Radius A is 6.52 m
12. A small clamp of mass m_B is attached at B to a hoop of mass m_h . The system is released from rest and rolls without sliding. Knowing that $m_h = 3m_R$. When $\theta = 90^\circ$



- (A) The angular acceleration of the hoop is $g/8r$
- (B) The angular acceleration of the hoop is $g/4r$
- (C) The horizontal components of the acceleration of B is $g/8$
- (D) The vertical components of the acceleration of B is $g/8$

Paragraph for Question Nos. 13 to 14

A small block of mass 1 kg starts moving with constant velocity 2 m/s on a smooth long plank of mass 10 kg which is also pulled by a horizontal force $F = 10t$ N where t is in seconds and F is in newton. (the initial velocity of the plank is zero).



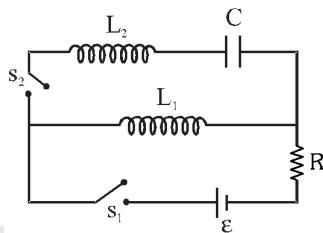
- 13.** Choose the correct option(s)

- (A) Displacement of 1 kg block with respect to plank at the instant when both have same velocity is $\frac{8}{3}$ m
- (B) Displacement of 1 kg block with respect to plank at the instant when both have same velocity is 4m
- (C) The time ($t \neq 0$) at which displacement of block and plank with respect to ground is same will be $2\sqrt{3}$ second :
- (D) The time ($t \neq 0$) at which displacement of block and plank with respect to ground is same will be 2 second:

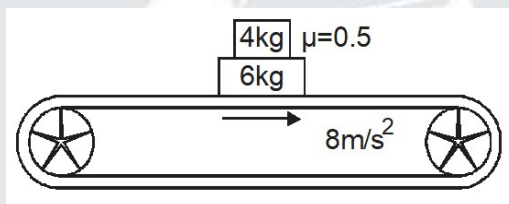
- 14.** Choose the correct option(s)

- (A) Relative velocity of plank with respect to block when acceleration of plank is 4 m/s^2 will be 6 m/s.
- (B) Relative velocity of plank with respect to block when acceleration of plank is 4 m/s^2 will be 4 m/s.
- (C) Acceleration of plank is 10 m/s^2 .
- (D) Acceleration of plank is variable.

15. Diagram shows a circuit. Switch S_1 was closed for a long time with S_2 open. At $t = 0$ switch S_2 was closed and S_1 was opened. The time period of L-C oscillation after $t = 0$ is given by $2\pi a \times 10^{-4}$ s. Fill the value of a . (Given $C = 30\mu\text{F}$; $L_1 = L_2 = 6\text{ mH}$, $R = 10\Omega$, $\varepsilon = 10\text{Volts}$).

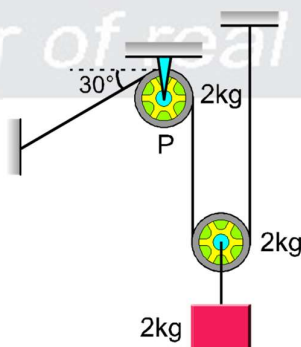


16. A system of two blocks is placed on an accelerating conveyor belt. If there is no slipping between 6kg block and conveyor belt and friction coefficient between 4kg & 6kg block is 0.5 then frictional force acting between 6kg & conveyor belt will be



17. A μ -meson particle moves in a circular orbit around a very heavy nucleus (of infinite mass) of charge $+3e$. Assuming Bohr's model is applicable to this system, find wavelength of radiation emitted in picometer when μ -meson jumps from 3rd orbit to 1st orbit [μ -meson is a particle, whose charge is equal to that of an electron and mass = 208 times that of an electron].

18. In the figure shown, mass of each pulley = 2 kg and mass of the block is also 2 kg. The strings are massless and pulleys are smooth. If the load (reaction) on the bearing of pulley P is $\sqrt{x}$ kgf then value of x is :



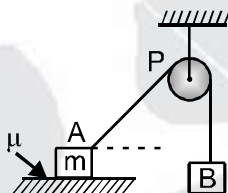
19. In a certain hypothetical radioactive decay process, species A decays in to species B and species B decays in to species C according to the reactions:



The decay constant for species A is λ_1 and that for species B is λ_2 . Initially there was an unknown amount of species A while there was none of B and C. It was found that species B reaches its maximum number at a time t_0 . $N \times 1,000$ moles of species A were there initially. Find N.

Take $\lambda_1 = 1 \text{ dps}$, $\lambda_2 = 100 \text{ dps}$, $N_{B_{\max}} = 2 \text{ moles}$, $t_0 = 21 \ln 10 \text{ sec}$.

20. In figure shown minimum mass of block B (at a particular angle between horizontal and string AP) to just slide the block A on rough horizontal surface is $\frac{m}{2}$ as shown in figure. If μ is the coefficient of friction between block A and ground then $\frac{1}{\mu^2}$ will be :



ANSWER KEY

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|-------------|-----------|-------------|----------|----------|
| 1. (B) | 2. (A) | 3. (C) | 4. (C) | 5. (ABC) |
| 6. (A) | 7. (C) | 8. (D) | 9. (AC) | 10. (AD) |
| 11. (AC) | 12. (ACD) | 13. (AC) | 14. (AD) | 15. (6) |
| 16. (68.00) | 17. (55) | 18. (28.00) | 19. (10) | |
| 20. (03.00) | | | | |